

Intuitionistic logic

- A logic without the Law of the Excluded Middle
-

In high school, we saw

Classical Logic (defined by truth table):

A proposition P, Q, R is

- either a propositional variable or connectives ($\neg, \wedge, \vee, \rightarrow$) of propositions,
- has a truth value of either true T or false F .

The truth value of connectives is defined by the truth table below:

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$
T	T	F	T	T	T
T	F	F	F	T	F
F	T	T	F	T	T
F	F	T	F	F	T

How to prove $\neg(P \vee Q) \rightarrow (\neg P \wedge \neg Q)$?

Just compute...

P	Q	$P \vee Q$	$\neg(P \vee Q)$	$\neg P$	$\neg Q$	$\neg P \wedge \neg Q$	$\neg(P \vee Q) \rightarrow (\neg P \wedge \neg Q)$
T	T	T	F	F	F	F	T
T	F	T	F	F	T	F	T
F	T	T	F	T	F	F	T
F	F	F	T	T	T	T	T

However, it is *NOT* how we prove a theorem!

To prove an implication $\neg(P \vee Q) \rightarrow (\neg P \wedge \neg Q)$,
we introduce the premise $\neg(P \vee Q)$ into our context, and try to derive the conclusion $\neg P \wedge \neg Q$.

To prove a conjunction $\neg P \wedge \neg Q$,
we need to derive the two subgoals $\neg P$ and $\neg Q$ separately under the same context.

1. To prove a negation $\neg P$,
we need to introduce the proposition P into our context, and try to derive a contradiction.

current proof state
context : $\neg(P \vee Q), P$
goal: contradiction

Since P holds, so $P \vee Q$ holds.
But now we have $\neg(P \vee Q)$ and $P \vee Q$, it is a contradiction.

2. To prove a negation $\neg Q$,
we need to introduce the proposition Q into our context, and try to derive a contradiction.

current proof state
context : $\neg(P \vee Q), Q$
goal: contradiction

Since Q holds, so $P \vee Q$ holds.
But now we have $\neg(P \vee Q)$ and $P \vee Q$, it is a contradiction.

We get $\neg P$ and $\neg Q$ under the context $\neg(P \vee Q)$, so $\neg P \wedge \neg Q$ hold under the context $\neg(P \vee Q)$.
We get $\neg P \wedge \neg Q$ hold under the context $\neg(P \vee Q)$, so $\neg(P \vee Q) \rightarrow (\neg P \wedge \neg Q)$ under the "empty context". ■

Assume $\neg(P \vee Q)$ hold.
If P hold, then $P \vee Q$, a contradiction.
If Q hold, then $P \vee Q$, a contradiction.
Therefore, $\neg P \wedge \neg Q$. ■

Key idea

At a moment of a proof, we have a *proof state*, consisting of

- context: what we know,
- goal: what we want to prove.

An *inference rule* is a transition between proof states.

We use inference rules to define what connectives mean, rather than using truth tables!!!

A *proposition* defines the grammar of a sentence.

Definition (Prop):

$$\begin{aligned} \text{PV } A, B &::= A \mid B \mid C \mid \dots \\ \text{Prop } P, Q &::= A \mid \perp \mid P \wedge Q \mid P \vee Q \mid P \rightarrow Q \end{aligned}$$

In other words:

- Let $\text{PV} = \{A, B, C, \dots\}$ be an infinite set of propositional variables.
- Let Prop be the set of all propositions, **inductively** defined by the following rules:
 - A is in Prop for all A in PV ,
 - $\perp$ is in Prop , and
 - $P \wedge Q, P \vee Q, P \rightarrow Q$ are in Prop , for all P, Q in Prop .

Abbreviation: $\neg P$ is a short hand of $P \rightarrow \perp$.

Convention: We implicitly add parentheses, e.g. $\neg P \vee \neg Q$ means $(\neg P) \vee (\neg Q)$, ...

Example: $\neg(P \vee Q) \rightarrow (\neg P \wedge \neg Q)$ is a proposition.

A *judgment* formalizes a proof state.

Definition (Judgment)

$$\begin{aligned} \text{Context } \Gamma &::= \cdot \mid \Gamma, P \\ \text{Judgment } J &::= \Gamma \vdash P \end{aligned}$$

In other words:

- A context Γ is a comma-separated finite list of propositions (including the empty list " $\cdot$ ").
- A judgment is a pair of a context Γ and a proposition P .

We pronounce $\Gamma \vdash P$ as " Γ entail P ".

Example: $\neg(P \vee Q) \rightarrow (\neg P \wedge \neg Q)$ is a proposition, and we write the judgment

$$\vdash \neg(P \vee Q) \rightarrow (\neg P \wedge \neg Q)$$

to say we want to prove this proposition without assuming anything.

An *inference rule* formalizes a transition from zero or more proof states to another proof state.

Definition (inference rule)

$$\frac{J_0 \quad J_1 \quad \dots \quad J_{n-1}}{J}(\text{name})$$

The judgments above/below the line are called premises/conclusion.

How to use inference rules:

- Instantiate: Substitute variables with specific contexts or propositions.
- Compose: Stack matching inference rules to form a proof tree.

Example: We will see an inference rule:

$$\frac{\Gamma, P \vdash Q}{\Gamma \vdash P \rightarrow Q}(\rightarrow I)$$

This inference rule can be instantiated to (take Γ be $\cdot$, P be $\neg(P \vee Q)$, Q be $\neg P \wedge \neg Q$)

$$\frac{\neg(P \vee Q) \vdash \neg P \wedge \neg Q}{\vdash \neg(P \vee Q) \rightarrow (\neg P \wedge \neg Q)}(\rightarrow I)$$

That is, to prove an implication $\neg(P \vee Q) \rightarrow (\neg P \wedge \neg Q)$,

we introduce the premise $\neg(P \vee Q)$ into our context, and try to derive the conclusion $\neg P \wedge \neg Q$.

Starting from a judgment, we will stack inference rules on it to build a proof tree until no premises remain.

The resulting proof tree is called a *derivation* of the judgment.

A *proof* of a proposition P is a derivation of the judgment $\vdash P$.

Example: The proof tree is a derivation of the judgment $\vdash \neg(P \vee Q) \rightarrow (\neg P \wedge \neg Q)$.

$$\frac{\frac{\frac{\vdash \neg(P \vee Q), P \vdash \neg(P \vee Q)}{\vdash \neg(P \vee Q), P \vdash \neg(P \vee Q)}(A) \quad \frac{\frac{\vdash \neg(P \vee Q), P \vdash P}{\vdash \neg(P \vee Q), P \vdash P \vee Q}(\vee I) \quad \frac{\vdash \neg(P \vee Q), P \vdash \perp}{\vdash \neg(P \vee Q), P \vdash \neg P}(\rightarrow I)}{\vdash \neg(P \vee Q) \vdash \neg P}(\rightarrow I) \quad \frac{\frac{\frac{\vdash \neg(P \vee Q), Q \vdash \neg(P \vee Q)}{\vdash \neg(P \vee Q), Q \vdash \neg(P \vee Q)}(A) \quad \frac{\vdash \neg(P \vee Q), Q \vdash Q}{\vdash \neg(P \vee Q), Q \vdash P \vee Q}(\vee I)}{\vdash \neg(P \vee Q), Q \vdash \neg Q}(\rightarrow I)}{\vdash \neg(P \vee Q) \vdash \neg Q}(\wedge I) \quad \frac{\vdash \neg(P \vee Q) \vdash \neg P \wedge \neg Q}{\vdash \neg(P \vee Q) \rightarrow (\neg P \wedge \neg Q)}(\rightarrow I)$$

Recap

- Build propositions using connectives.
- Form a judgment by pairing a context with a goal proposition.
- Compose inference rules to derive a judgment.

A *natural deduction* system is a collection of inference rules.

The **intuitionistic** variant we are introducing is named NJ by [Gentzen](#).

Assumption rule:

$$\frac{}{\Gamma \vdash P} (A)$$

with the *side condition* that P appear in Γ .

For each connective, we need to define how to construct it and how to use it.

If a connective appears in a premise/conclusion of an inference rule, this rule is called an *introduction/elimination* rule.

Conjunction:

- Introduction:

$$\frac{\Gamma \vdash P \quad \Gamma \vdash Q}{\Gamma \vdash P \wedge Q} (\wedge I)$$

- Elimination:

$$\frac{\Gamma \vdash P \wedge Q}{\Gamma \vdash P} (\wedge EL) \quad \frac{\Gamma \vdash P \wedge Q}{\Gamma \vdash Q} (\wedge ER)$$

Exercise: Derive $(P \wedge Q) \wedge R \vdash P \wedge (Q \wedge R)$

Use Obsidian Canvas: [Exercise Blackboard.canvas](#)

Implication:

- Introduction:

$$\frac{\Gamma, P \vdash Q}{\Gamma \vdash P \rightarrow Q} (\rightarrow I)$$

- Elimination:

$$\frac{\Gamma \vdash P \rightarrow Q \quad \Gamma \vdash P}{\Gamma \vdash Q} (\rightarrow E)$$

Exercise: Derive $(P \wedge Q) \rightarrow R \vdash P \rightarrow (Q \rightarrow R)$

Disjunction:

- Introduction:

$$\frac{\Gamma \vdash P}{\Gamma \vdash P \vee Q} (\vee\text{IL}) \quad \frac{\Gamma \vdash Q}{\Gamma \vdash P \vee Q} (\vee\text{IR})$$

- Elimination:

$$\frac{\Gamma \vdash P \vee Q \quad \Gamma, P \vdash R \quad \Gamma, Q \vdash R}{\Gamma \vdash R} (\vee\text{E})$$

Exercise: Derive $(P \vee Q) \rightarrow R \vdash (P \rightarrow R) \wedge (Q \rightarrow R)$

Falsity:

- Introduction: none
- Elimination:

$$\frac{\Gamma \vdash \perp}{\Gamma \vdash P} (\perp\text{E})$$

Exercise: Derive $P \vdash \neg\neg P$

Question: What is the difference between $\rightarrow, \vdash, \frac{J}{J'}$?

More precisely, do the following inference rules *imply* each other under NJ?

$$\overline{\vdash P \rightarrow Q} \quad \overline{\Gamma \vdash P \rightarrow Q} \quad \overline{P \vdash Q} \quad \overline{\Gamma, P \vdash Q} \quad \frac{\Gamma \vdash P}{\Gamma \vdash Q}$$

We say an inference rule (R_1) imply (R_2) under a natural deduction N system if

One can replace any occurrences of the rule (R_1) in a proof tree built by $N \cup \{(R_1)\}$ by a sub proof tree built by $N \cup \{(R_2)\}$.

see: [Exercise Blackboard.canvas](#) (using *weakening rule*)

We can prove	but not	
$\neg\neg(P \vee \neg P)$	$P \vee \neg P$	law of excluded middle
$P \rightarrow \neg\neg P$	$\neg\neg P \rightarrow P$	double negation
$(\neg P \wedge \neg Q) \leftrightarrow \neg(P \vee Q)$		De Morgan's laws
$(\neg P \vee \neg Q) \rightarrow \neg(P \wedge Q)$	$\neg(P \wedge Q) \rightarrow (\neg P \vee \neg Q)$	De Morgan's laws
$(P \rightarrow Q) \rightarrow (\neg Q \rightarrow \neg P)$	$(\neg Q \rightarrow \neg P) \rightarrow (P \rightarrow Q)$	contraposition
$(\neg P \vee Q) \rightarrow (P \rightarrow Q)$	$(P \rightarrow Q) \rightarrow (\neg P \vee Q)$	oh?

Proof of "We can prove": [Exercise Blackboard.canvas](#)

Remarkably, in intuitionistic logic NJ:

- There are not concepts about truth value, and so the law of excluded middle and double negation are not trivial.
- Disjunction seems more stronger then in classical logic, it requires a explicit proof of P or a proof of Q .

Question: How to prove these proposition P in "but not" are not provable i.e. $\not\vdash P$.

The Law of The Excluded Middle

The classical natural deduction NK is defined by NJ adding the law of excluded middle (LEM) or adding the double negation ($\neg\neg$ E)

$$\frac{}{\Gamma \vdash P \vee \neg P}(\text{LEM}) \quad \frac{}{\Gamma \vdash \neg\neg P \rightarrow P}(\neg\neg\text{E})$$

Question: Why does adding one of the two rules suffice?

We say two inference rules $(R_1), (R_2)$ are *equivalent* under a natural deduction N system if

One can replace any occurrences of rule (R_1) in a proof tree build by $N \cup \{(R_1)\}$ by a sub proof tree build by $N \cup \{(R_2)\}$, and vice versa.

Exercise: Under NJ, the above two rule (LEM), ($\neg\neg$ E) also equivalent to the following two rules

$$\frac{\Gamma, P \vdash Q \quad \Gamma, \neg P \vdash Q}{\Gamma \vdash Q}(\text{by cases}) \quad \frac{\Gamma, \neg P \vdash \perp}{\Gamma \vdash P}(\text{indirect proof})$$

Exercise: also equivalent to

$$\frac{\Gamma \vdash P \rightarrow Q}{\Gamma \vdash \neg P \vee Q}(\text{oh?})$$

Answer: [Exercise Blackboard.canvas](#)

!!!

[You just assumed the Law of the Excluded Middle](#)

Classical logic (defined by natural deduction) vs Classical logic (defined by truth table)

syntactic $\Gamma \vdash P$ vs semantic $\Gamma \models P$

see: [Exercise Blackboard.canvas](#)

Disjunction Property

Why can't we derive $\Gamma \vdash P \vee \neg P$?

Theorem: (disjunction property):

Given a proof tree of $\vdash_{\text{NJ}} P_0 \vee P_1$, we can compute an index $i \in \{0, 1\}$ along with a proof tree of $\vdash_{\text{NJ}} A_i$.

Question: What is the difference between the statement of the above theorem and

"If $\vdash_{\text{NJ}} P \vee Q$, then either $\vdash_{\text{NJ}} P$ or $\vdash_{\text{NJ}} Q$."

Remark: The disjunction property does not hold for NK.

Question: Is the above remark related to the fact that the CH (Continuum Hypothesis) is independent of ZFC (Zermelo-Fraenkel set theory with the axiom of choice)?

Gödel, 1940: $\text{ZFC} \not\vdash \text{CH}$

Cohen, 1963: $\text{ZFC} \not\vdash \neg \text{CH}$

Law of Excluded Middle: $\text{ZFC} \vdash \text{CH} \vee \neg \text{CH}$

Constructive Logic

finite vs infinite

Abbreviation:

Let P decidable be a shorthand for $P \vee \neg P$

Let Γ decidable be a shorthand for P_0 decidable, P_1 decidable, ..., P_{n-1} decidable

- when Γ is $P_0, P_1, \dots, P_{n-1}$

In NJ, we can derive

$$\begin{array}{ll} P \text{ decidable,} & \neg(\neg P) \vdash P \\ P, Q \text{ decidable,} & \neg(\neg P \wedge \neg Q) \vdash P \vee Q \\ P, Q, R \text{ decidable,} & \neg(\neg P \wedge \neg Q \wedge \neg R) \vdash P \vee Q \vee R \\ & \vdots \\ \Gamma \text{ decidable,} & \neg(\bigwedge_{P \in \Gamma} \neg P) \vdash \bigvee_{P \in \Gamma} P \end{array}$$

De Morgan's laws are allowed when the collection of propositions is *finite* and *decidable*.

In NK, all propositions are decidable.

But more importantly, the law of excluded middle allows De Morgan's laws in an *infinite* setting.

Let prove $\neg(\forall x. \neg P(x)) \vdash_{\text{NK}+\forall\exists} \exists x. P(x)$ by natural language.

Assume $\neg(\forall x. \neg P(x))$ holds.

By indirect proof, assume toward contradiction that $\neg(\exists x. P(x))$ holds.

- Claim: $\forall x. \neg P(x)$

Let x be such $P(x)$ holds.

Then we find a x such that $P(x)$. So $\exists x. P(x)$ holds. Contradiction.

We get $\forall x. \neg P(x)$, it contradiction to $\neg(\forall x. \neg P(x))$ ■.

Question: Can we prove $\forall x. (P(x) \text{ decidable}), \neg(\forall x. \neg P(x)) \vdash_{\text{NJ}+\forall\exists} \exists x. P(x)$?

After this talk, you know

- How to use natural deduction (grammar, judgment, inference rule) to define a logic system.
- How to use inference rules to build a derivation of a proposition.
- You have a choice to accept the law of excluded middle or not.



Reference:

2026 flolac 課程 邏輯

<https://flolac.iis.sinica.edu.tw/zh/2026/>

https://flolac.iis.sinica.edu.tw/2026/logic_lecture_I.pdf

https://flolac.iis.sinica.edu.tw/2026/logic_lecture_IV.pdf

Natural Deduction for Intuitionistic Logic | Attic Philosophy

https://youtu.be/7pmQkGLlqY0?si=NUmSKdZzSG6_y60p

Further reading: (I want to read it, but have no time, please, read it and teach me)

Constructive Analysis

<https://link.springer.com/book/10.1007/978-3-642-61667-9>

On intuitionistic linear logic

<https://www.cl.cam.ac.uk/techreports/UCAM-CL-TR-346.html>

From Classical to Intuitionistic Probability, Brian Weatherson

<https://brian.weatherson.org/conprob.pdf>

羅翔 bilibili 疑罪从无

[https://www.bilibili.com/video/BV1DEcszgED9/?](https://www.bilibili.com/video/BV1DEcszgED9/?share_source=copy_web&vd_source=d3096126491cbef625f1873051b0f078)

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被告否認犯罪事實所持之辯解，縱屬不能成立，仍非有積極證據足以證明其犯罪行為，不能遽為有罪之認定。

[https://mojlaw.moj.gov.tw/LawContentExShow.aspx?](https://mojlaw.moj.gov.tw/LawContentExShow.aspx?media=print&id=B,30,%E4%B8%8A,1831,001&type=J1)

[media=print&id=B,30,%E4%B8%8A,1831,001&type=J1](https://mojlaw.moj.gov.tw/LawContentExShow.aspx?media=print&id=B,30,%E4%B8%8A,1831,001&type=J1)